

SAHAJ DEEP SINGH

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Experience

ESDS Software Solution

June 2025 – Present

Research and Development Associate

Nashik, Maharashtra

- Architected a secure, distributed remote execution system to manage parallel command orchestration across **100+** multi-agent server environments using **Python**, **FastAPI**, **gRPC**, and **asyncio**.
- Led the backend production migration to a high-performance, event-driven microservices architecture in **Go**, utilizing **gRPC** and **Redis Pub/Sub** to handle high-volume data streams securely.
- Automated the creation and configuration of virtual machines using **QEMU**, utilizing **cloud-init** scripts to handle dynamic package initialization and automated OS bootstrapping on initialization.
- Secured and hardened a multi-tenant **Kubeflow** cloud environment, deploying an **Istio service mesh** and fine-grained **RBAC** network policies to completely isolate shared computing workloads.
- Orchestrated the secure network integration of **5+** third-party software applications into the core enterprise platform by managing ingress routing, external APIs, and access control policies.
- Streamlined infrastructure-as-code deployment pipelines using modular **Kustomize** manifests and **Shell** automation to completely eliminate environment configuration drift across clusters.
- Developed an internal AI-driven data triage system using **Python** and **scikit-llm** to automatically parse, classify, and route **800+** daily infrastructure system alerts, successfully eliminating manual sorting queues.

Mahindra and Mahindra

Feb 2024 – Mar 2024

Software Engineering Intern

Chennai, Tamil Nadu

- Built a vehicle headlight illumination prediction model using data analytics and **linear regression** to estimate physical road visibility boundaries based on luminous intensity and beam deflection angles.

Technical Skills

Languages: Go, Python, Java, Bash, JavaScript

AI Engineering: LangGraph, LangChain, MCP, LLMs, NLP, Sentence Transformers, ONNX Runtime, scikit-llm

Backend Engineering: FastAPI, Flask, gRPC, REST APIs

Architecture & Concepts: Distributed Systems, Microservices, Event-Driven Architecture, GitOps

Infra & Platform: Kubernetes, Docker, Istio

Databases & Messaging: PostgreSQL, Redis, MySQL, SQLite

Developer Tools: Git, Linux, QEMU, Neovim, Tmux, Alacritty

Projects

Orgbrain | Enterprise AI Privacy Interceptor & Knowledge Bus | *Python, FastAPI, PostgreSQL, pgvector* [Code](#)

- Developed an enterprise privacy and knowledge platform that stops confidential data and credentials from leaking into external AI models, while enabling engineering teams to safely share and discover solutions across sessions.
- Architected an interception gateway that transparently inspects AI prompt payloads to strip credentials and personal data, retrieves relevant past developer sessions, and enforces human-in-the-loop review before any data is saved.
- Engineered an asynchronous pipeline using **FastAPI**, **PostgreSQL**, and **pgvector** with on-premise **FastEmbed** embeddings, streaming real-time token restoration over **SSE** and enforcing write idempotency validated by **140+ tests**.

Semantic Interview Evaluation Engine | *Python, Flask, ONNX Runtime, NLP, Docker* [Code](#)

- Developed an asynchronous, containerized platform to automate video-based technical interview evaluations, transcript generation, and semantic grading.
- Automated precise transcript grading and minimized evaluation latency by exporting open-source **NLP models** into an optimized **ONNX Runtime** engine for vector similarity scoring.
- Guaranteed clean production deployments and seamless database schema updates by containerizing the stack with **Docker/Gunicorn** and integrating **SQLAlchemy** migrations.

JScraper | Concurrent Job Aggregation Pipeline | *Go, Python, SQLite, Sentence Transformers* [Code](#)

- Developed a high-throughput, open-source web crawling and data aggregation tool in **Go** designed to ingest, filter, and process multi-domain job listings.
- Prevented target-domain IP bans and optimized network resource consumption by architecting an extraction loop with token-bucket rate limiting and URL deduplication.
- Eliminated redundant network fetches and process crashes across scheduled batch cycles by implementing layered fault tolerance via a **SQLite**-persisted circuit breaker and exponential backoff.

Education

SRM Institute of Science and Technology, Chennai, India

Sep 2021 – Jun 2025

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.86/10